A Unified Effective Mechanism for Galaxy Rotation Curves and the Hubble Tension

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ABSTRACT

We investigate whether two long-standing observational anomalies—flat galaxy rotation curves and the Hubble constant tension—may originate from a common effective mechanism operating in low-density environments.

We introduce a minimal phenomenological framework in which departures from Newtonian expectations arise from a saturation of the effective gravitational response at the level of kinematic inference, without invoking dark matter particles or altering early-universe cosmology. In the galactic regime, the well-known MOND phenomenology emerges as the low-density limit of this bounded-response framework, rather than as a fundamental modification of gravity. The framework reduces to standard Newtonian behavior in high-density regimes and introduces a smooth, bounded effective response in diffuse environments.

Applied to galactic dynamics, this approach reproduces the main features of observed rotation curves across a wide range of galaxy types using baryonic matter alone, with stable parameter values and no halo-by-halo tuning. Explicit numerical comparisons with representative galaxies from the SPARC sample show that the effective saturation scale naturally correlates with observed surface-density-dependent trends.

At cosmological scales, the same effective mechanism predicts environment-dependent deviations in the locally inferred expansion rate. Cosmic voids emerge as maximal probes of the saturated regime, leading to a systematic offset between local and global measurements of the Hubble constant, while preserving the background cosmological evolution. This provides a structural interpretation of the Hubble tension without introducing new dark components or modifying early-time physics.

We examine robustness, degeneracies with standard astrophysical effects, and limitations of the effective description, and we outline distinctive observational signatures, including environment-dependent redshift drift and weak lensing effects. The results suggest that galaxy rotation curves and the Hubble tension may reflect a shared low-density phenomenology, testable with current and forthcoming observations.

Key words: galaxies: kinematics and dynamics – cosmology: observations – cosmology: theory – large-scale structure of Universe

1 INTRODUCTION

Two observational puzzles continue to motivate scrutiny of late-time gravitational phenomenology.

The first is the ubiquity of flat galaxy rotation curves. Across a wide range of disk galaxies, circular velocities remain nearly constant at large radii, in apparent tension with the expectation from Newtonian dynamics applied to the observed baryonic mass distribution. This behavior has been firmly established by large and homogeneous datasets, most notably the SPARC compilation, which provides resolved rotation curves together with self-consistent baryonic mass models for disk galaxies Lelli et al. (2016). Within the standard cosmological framework, flat rotation curves are typically accounted for by extended dark matter halos. While this approach is empirically successful, it introduces substantial model freedom at the level of individual galaxies and leaves open the question of why tight empirical regularities, such as surface-density-dependent trends and acceleration-based scalings, emerge so prominently in the data McGaugh et al. (2016). Historically, these regularities led to the development of the Modified Newtonian Dynamics (MOND) paradigm Milgrom (1983); Bekenstein & Milgrom (1984), which remains the primary phenomenological benchmark for such anomalies.

The second puzzle is the Hubble constant tension. Late-time local determinations of H_0 based on distance ladder techniques and standard candles remain in significant disagreement with early-time inferences derived from the homogeneous high-redshift Universe as constrained by cosmic microwave background observations Riess (2022); Collaboration (2020). This discrepancy has persisted across multiple datasets and analysis methods, prompting proposals ranging from unidentified systematics to extensions of the standard cosmological model. Comprehensive reviews indicate that no consensus explanation has yet emerged Verde et al. (2019). A notable feature of the tension is its strong association with late-time inference in a structured and inhomogeneous Universe.

Although galaxy rotation curves and the Hubble tension are usu-

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ally discussed separately, both phenomena probe kinematic inference in low-density regimes. Galaxy outskirts and ultra-diffuse systems explore weak-acceleration regions, while cosmic voids represent the most extreme low-density environments on cosmological scales. This commonality motivates the central question addressed in this work. Can a single effective mechanism, active primarily in diffuse environments, provide a consistent phenomenological interpretation of both classes of anomalies within a unified descriptive framework.

In this paper, we adopt a perspective in which late-time cosmological observables are interpreted as effective quantities arising from a coarse-grained description of admissible macroscopic states. Within this view, quantities such as gravitational acceleration or the Hubble parameter are not treated as fundamental dynamical variables. They characterize effective transition rates between macrostates of the observable Universe under a given resolution of description.

Departures from Newtonian expectations in low-density environments are then not attributed to a modification of the force law, but to a limitation of the effective kinematic inference applied to increasingly diffuse configurations. They reflect a saturation of the effective description when further refinement of the inferred response becomes impossible. In this regime, the effective gravitational response does not continue to grow indefinitely as density decreases, but instead approaches a plateau controlled by an intrinsic saturation scale.

While the mathematical structure of this saturation in the galactic limit shares similarities with the a-field theories of the Bekenstein–Milgrom formulation Bekenstein & Milgrom (1984), the present approach is conceptually distinct and does not posit a modification of the underlying equations of motion. The saturation does not represent an ad hoc modification of gravity. It arises from a limitation of the effective description applicable in diffuse regimes, where kinematic inference becomes progressively insensitive to further reductions in density.

This phenomenological study is motivated by a broader theoretical context in which bounded-response regimes of the Born–Infeld type arise naturally in effective descriptions of gravitational dynamics Beau (2026a). Related structural and projection-based considerations are discussed in companion works focusing on the informational foundations of the geometric support Beau (2026b,c). Here we deliberately restrict attention to late-time observational consequences and do not rely on the full underlying framework.

The model is constructed to reduce to standard Newtonian behavior in high-density environments and to introduce a smooth bounded modification in diffuse regimes. It is treated explicitly as an effective description of late-time phenomenology. No modification of early-universe physics is assumed, and the model is not presented as a fundamental theory of gravity.

The analysis proceeds in two stages. First, we test the model against galaxy rotation curve data, focusing on the SPARC sample, and evaluate whether a single universal saturation scale can reproduce the observed diversity of rotation curve shapes using baryonic matter alone, without halo-by-halo tuning. Second, we explore the cosmological implications of the same mechanism in low-density environments, emphasizing cosmic voids as maximal probes and deriving testable signatures for environment-dependent expansion inference.

A key aim of this work is falsifiability. Beyond fitting existing data, we identify observational discriminants that can confirm or exclude the framework. These include correlations between locally inferred values of H_0 and large-scale environment, redshift-dependent suppression of the local offset, void-specific redshift drift, and weak lensing signatures.

The structure of the paper is as follows. Section 2 introduces the effective model and its minimal assumptions. Section 3 presents rotation curve tests and diagnostics. Section 4 discusses the implications for cosmic voids and the Hubble tension. Section 5 examines robustness, degeneracies, and limits. Section 6 summarizes predictions and observational tests. Section 7 concludes.

2 EFFECTIVE MODEL

This section introduces a minimal effective framework designed to capture late-time gravitational phenomenology in low-density environments. The model is constructed to address galaxy rotation curves and the Hubble tension within a single unified description, without modifying early-universe physics or invoking dark matter particles in this effective regime.

The emphasis is deliberately phenomenological. Only the minimal assumptions required to reproduce the observed anomalies are introduced. The model is not presented as a fundamental theory of gravity, but as an effective description valid in specific environmental regimes.

The guiding interpretation adopted here is that late-time gravitational observables characterize effective transition rates between admissible macroscopic configurations of the observable Universe. In this view, quantities such as gravitational acceleration or the Hubble parameter do not represent fundamental dynamical fields. They encode how rapidly the effective state of the Universe can be updated under a finite resolution of the underlying descriptive structure.

Saturation is interpreted as a limitation of this effective resolution. In low-density environments, the number of independent relations contributing to the gravitational response becomes bounded, so that further decreases in density no longer translate into proportionally stronger inferred dynamics. As a result, the effective response approaches a plateau rather than vanishing asymptotically. The dynamical motivation for bounded-response regimes of this type is developed in a companion theoretical work Beau (2026a).

2.1 Minimal Assumptions and Domain of Validity

We assume that gravitational phenomenology at galactic and subcosmological scales can be described by an effective potential Φ_{eff}, used here as a convenient parametrization of kinematic inference rather than as a fundamental dynamical field. The framework satisfies the following requirements.

(i) The effective dynamics must reduce to standard Newtonian gravity in high-density environments, ensuring consistency with Solar System tests. (ii) Deviations are allowed only in diffuse regimes, characterized by low baryonic density or weak gravitational gradients. (iii) The transition is governed by at most one additional universal scale parameter $a_\star$, with no environment-specific tuning. (iv) Locality and isotropy are preserved at the effective level.

Mathematically, the framework shares the Lagrangian structure of the quasi-linear MOND theories proposed by Bekenstein and Milgrom Bekenstein & Milgrom (1984). However, whereas MOND introduces this modification as a fundamental alteration of the gravitational law, the present approach interprets it as a saturated effective response. The modification reflects a finite capacity of the effective description to resolve increasingly dilute relational structure Beau (2026c).

This conceptual shift is essential. It allows the phenomenological success of MOND at galactic scales to be reinterpreted as a regime of descriptive saturation, rather than as a modification of gravity. The same mechanism then applies naturally to volumetric density dilution in cosmological voids.

The present framework is therefore not intended as a replacement for MOND phenomenology. It provides an interpretation of its low-acceleration successes as an emergent effective limit within a broader descriptive structure.

2.2 Effective Saturating Potential

The field equation introduced below is formally identical to the quasi-linear MOND formulation in the deep galactic regime. In the present work, however, this structure is not postulated as a fundamental modification of gravity. Instead, we adopt it as a minimal effective parametrization of a bounded-response principle governing low-density kinematic inference.

We define the effective gravitational potential $\Phi_{\rm eff}$ through the nonlinear Poisson equation

$$\nabla \cdot \left[\mu\left(\frac{|\nabla\Phi_{\rm eff}|}{a_\star}\right)\nabla\Phi_{\rm eff}\right] = 4\pi G\rho_{\rm b}, \qquad (1)$$

where $\rho_{\rm b}$ denotes the baryonic mass density. The function $\mu(x)$ encodes the progressive saturation of the effective response as the acceleration falls below the characteristic scale $a_\star$.

For clarity, we work primarily with the μ-formulation in Eq. (1). An equivalent algebraic representation may be written in terms of an interpolation function $\nu(y)$ defined by $g_{\rm eff} = \nu(y)g_{\rm N}$ with $y = g_{\rm N}/a_\star$, where $\nu(y) \to 1$ for $y \gg 1$, ensuring an exactly Newtonian limit.

In the present interpretation, $\mu(x)$ does not represent a field-dependent coupling in the fundamental action. Rather, it parametrizes the smooth onset of a bounded-response regime in the effective description. The specific functional form adopted here provides a minimal interpolation between regimes and is not assumed to be unique.

The function $\mu(x)$ satisfies

$$\mu(x) \to 1 \quad \text{for} \quad x \gg 1, \qquad (2)$$

and

$$\mu(x) \to x \quad \text{for} \quad x \ll 1, \qquad (3)$$

ensuring recovery of Newtonian dynamics at high acceleration and asymptotically flat rotation curves at low acceleration.

For spherically symmetric systems, the effective radial acceleration satisfies

$$\mu\left(\frac{g_{\rm eff}}{a_\star}\right) g_{\rm eff} = g_{\rm N}, \qquad (4)$$

where $g_{\rm N}$ is sourced solely by baryons.

Equation (4) reproduces the observed flattening of galaxy rotation curves without introducing dark halos. The transition scale $a_\star$ controls both the onset of flattening and the asymptotic velocity amplitude.

Crucially, $a_\star$ is not treated as a tunable per-galaxy parameter. It is determined globally from the full rotation-curve sample (Section 3.4), and is interpreted here as the characteristic acceleration at which the effective geometric response enters a bounded regime.

2.3 Environmental Dependence and Low-Density Regimes

The effective modification introduced above is intrinsically sensitive to the gravitational environment. Regions of low baryonic density or weak gravitational gradients probe the saturated regime most strongly.

This environmental dependence becomes critical at cosmological scales. Cosmic voids, characterized by extreme matter dilution, naturally sample the deep saturation regime. As a result, local kinematic measurements performed in void-dominated regions may exhibit systematic offsets relative to globally inferred expansion rates.

Within this framework, the Hubble tension is not interpreted as a failure of early-universe cosmology. It arises as a late-time inference bias, reflecting the use of a single spacetime-based parametrization across environments characterized by different effective resolutions.

Galaxy rotation curves and the Hubble tension therefore emerge as two manifestations of the same low-density phenomenology. Galaxies probe saturation radially through declining surface density, while cosmic voids probe it volumetrically through large-scale dilution. The identification of a single scale $a_\star$ accounting for both effects constitutes the central organizing principle of this effective description.

The following sections test this effective framework against galactic rotation curve data and explore its implications for local measurements of the Hubble constant.

2.4 Operational Saturation Scale

The bounded-response behavior introduced above can be associated with an operational saturation scale. In the underlying relational description, the invariant bound b_χ limits the maximal rate at which admissible configurations can be updated. In the effective macroscopic regime, this constraint manifests as a characteristic acceleration proxy $a_\star$ marking the onset of non-linear response.

Dimensional consistency requires that this threshold be inversely proportional to the descriptive resolution $\ell(z)$ of the effective state U. We therefore parametrize

$$a_\star(z) \sim \kappa \frac{b_\chi}{\ell(z)}, \qquad (5)$$

where κ is a dimensionless normalization factor encoding the convention adopted for the projective mapping. In the late-time low-density regime considered in this section, $a_\star$ may be treated as approximately constant. Its possible redshift dependence becomes relevant only in the primordial regime and will be discussed in Section 6.

3 GALAXY ROTATION CURVES

Throughout this section, the only galaxy-dependent free parameter is the stellar disc mass-to-light ratio $\Upsilon_\star$, varied within a physically motivated interval. The saturation scale $a_\star$ is determined once from the full sample and then held fixed. No additional halo parameters are introduced.

We now test the effective model introduced in Section 2 against observed galaxy rotation curves. The goal of this section is not to achieve maximal fitting flexibility, but to assess whether a single effective saturation scale can reproduce the main kinematic features of disk galaxies using baryonic matter alone.

A useful internal consistency requirement is that the acceleration scale governing the transition be determined directly from galactic kinematic data. For the SPARC sample, fitting the effective model to the observed rotation curves yields a characteristic transition scale of order $10^{-10}\,{\rm m\,s^{-2}}$. This value is obtained within the present framework solely from galaxy rotation data and does not rely on any cosmological prior or assumption regarding the Hubble constant.

The numerical value is consistent with the empirical acceleration scale identified in MOND analyses McGaugh et al. (2016). In the present framework, however, this scale is not introduced as a fundamental modification of the gravitational law. It emerges as the transition point of a bounded-response regime in the effective kinematic description.

This motivates treating $a_\star$ as a single universal effective scale consistently appearing in both high- and low-surface-density galaxies, without requiring modifications to early-time cosmology or additional dark components.

3.1 Data and Methodology

We use the SPARC database, which provides high-quality rotation curves together with homogeneous photometry and baryonic mass models for disk galaxies spanning a wide range of morphologies, surface brightnesses, and dynamical regimes Lelli et al. (2016).

The numerical pipeline used to generate the rotation-curve figures and diagnostics is available as open-source software Beau (2026d).

For each galaxy, the baryonic mass distribution is constructed from the observed stellar and gas components. Stellar mass-to-light ratios are fixed following the standard SPARC prescriptions. No additional dark matter component is introduced in this effective description.

The effective acceleration $g_{\rm eff}(r)$ is obtained by solving Eq. (4) using the Newtonian acceleration $g_{\rm N}(r)$ computed from the baryonic mass distribution. The circular velocity then follows from

$$v_{\rm eff}(r) = \sqrt{r\, g_{\rm eff}(r)}. \tag{6}$$

The saturation scale $a_\star$ is treated as a universal parameter. Its value is fixed globally by minimizing the total residuals across the sample, rather than optimized on a galaxy-by-galaxy basis.

3.2 Representative Fits

We illustrate the performance of the effective model on representative galaxies spanning different surface-density regimes. The goal is not to emphasize isolated best cases, but to show typical behaviors observed across the full SPARC sample.

High-surface-density galaxies are well described by Newtonian dynamics in their inner regions. Deviations from Newtonian predictions appear only at large radii, where the baryonic acceleration drops below the saturation scale. The resulting rotation curves flatten smoothly, without requiring extended mass halos, in agreement with observed trends across high-surface-brightness disks Lelli et al. (2016).

Low-surface-brightness galaxies probe the saturated regime over most of their radial extent. In these systems, the effective acceleration departs from Newtonian behavior already at small radii, naturally producing slowly rising and asymptotically flat rotation curves, a feature commonly observed in diffuse systems McGaugh et al. (2016).

Across the sample, the model reproduces the observed diversity of rotation curve shapes. This diversity arises solely from differences in the baryonic mass distribution and surface density, not from variations in the effective parameters.

The galaxies shown in Fig. 1 are selected as representative well-fitted systems spanning rising, nearly flat, and slowly declining rotation curves. The saturation scale $a_\star$ is fixed globally from the full-sample analysis discussed in Section 3.4. No galaxy-dependent adjustment of $a_\star$ is performed.

Representative examples illustrating these behaviors are shown in Fig. 1.

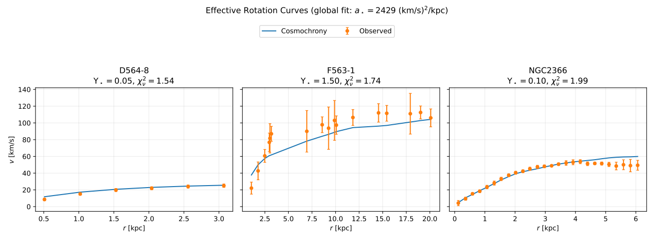

Figure 1. Representative galaxy rotation curves from the SPARC sample. Observed circular velocities (points) are compared with the effective model prediction (solid lines) computed from the baryonic mass distribution alone. The three systems span distinct surface-density regimes and kinematic behaviors (rising, nearly flat, and mildly declining). A single universal saturation scale $a_\star$, determined from the full sample, is used for all galaxies. The only galaxy-dependent parameter is the stellar mass-to-light ratio $\Upsilon_\star$, fitted within a physically motivated interval. The code used to generate these curves is publicly available (Beau 2026d).

3.3 Residuals and Scaling Relations

We quantify the quality of the fits using radial velocity residuals

$$\Delta v(r) = v_{\rm obs}(r) - v_{\rm eff}(r). \tag{7}$$

The residuals exhibit no systematic radial trends across the sample. In particular, no characteristic scale or radius-dependent bias is observed.

The effective model naturally reproduces the empirical correlation between rotation curve shape and baryonic surface density. Galaxies with higher central surface density remain in the Newtonian regime over a larger radial range, while diffuse systems enter the saturated regime earlier.

The transition radius at which rotation curves flatten corresponds closely to the regime where the effective surface flux density approaches the saturation limit. Beyond this point, further decreases in baryonic density no longer translate into stronger dynamical suppression, leading to asymptotically flat rotation velocities.

This behavior is closely related to the observed radial acceleration relation. In the present framework, this relation is not imposed but emerges directly from the effective saturation of the gravitational response at low acceleration McGaugh et al. (2016).

3.4 Universality of the Saturation Scale

A central outcome of the analysis is the empirical stability of the saturation scale $a_\star$ when determined from the full SPARC sample (N = 175 galaxies). The scale is obtained through a global robust optimization, varying only the stellar disc mass-to-light ratio $\Upsilon_\star$ on a galaxy-by-galaxy basis, while keeping Υ_{bul} fixed.

Using a median-based estimator and allowing $\Upsilon_\star \in [0.01, 2.0]$, we obtain $a_\star \simeq 2.4 \times 10^3\,(\mathrm{km\,s^{-1}})^2\,\mathrm{kpc}^{-1}$. A trimmed-mean criterion yields $a_\star \simeq 2.8 \times 10^3\,(\mathrm{km\,s^{-1}})^2\,\mathrm{kpc}^{-1}$. The inferred scale is therefore stable at the $\sim$ 15% level with respect to the choice of robust estimator.

The resulting distribution of reduced χ^2 values is shown in Fig. 2. The median value is $\chi^2_\nu \approx 2.7$, with approximately 60–65% of galaxies satisfying $\chi^2_\nu < 5$ and nearly 80% satisfying $\chi^2_\nu < 10$. These values are obtained without introducing galaxy-dependent halo parameters or cosmological priors.

The inferred scale remains within the same range when adopting plausible variations of the bulge mass-to-light ratio ($\Upsilon_{bul} = 0.3$–0.7),

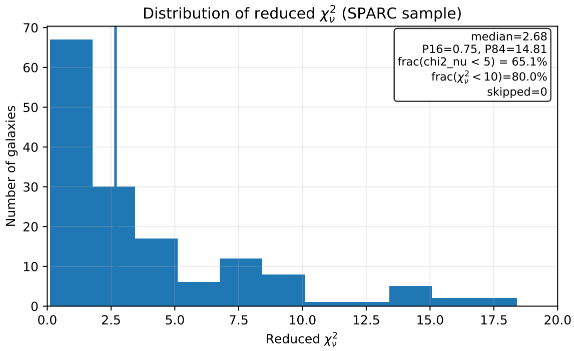

Figure 2. Distribution of reduced χ^2 values for the full SPARC sample (N = 175 galaxies) using the globally fitted saturation scale $a_\star$. The vertical line indicates the median value.

indicating that $a_\star$ is not driven by fine-tuning of baryonic normalization.

This supports the interpretation of $a_\star$ as a universal effective transition scale rather than as a phenomenological fitting parameter. All galaxy-to-galaxy variability is accounted for by the observed baryonic structure, with no halo-by-halo adjustment.

In the present framework, the universality of the saturation scale reflects the existence of a bounded-response regime in the effective geometric description, rather than a modification of the fundamental gravitational law.

The statistical performance obtained here is comparable to that reported in published MOND fits to the same SPARC dataset, but is derived from an effective saturation principle rather than a postulated modification of the gravitational field equation.

The absence of a secondary high-χ^2 population suggests that the model does not rely on a fine-tuned subset of galaxies.

3.5 Summary

The effective saturation model reproduces the main phenomenological features of galaxy rotation curves across a wide range of systems using baryonic matter alone. The same universal saturation scale governs both high- and low-surface-density galaxies, with smooth transitions between Newtonian and saturated regimes.

These results motivate extending the analysis to cosmological low-density environments. In the next section, we show that the same effective mechanism leads to environment-dependent signatures in the inferred expansion rate, providing a natural connection to the Hubble tension.

The determination of $a_\star$ from galactic rotation curves constitutes the only free parameter of the effective model. No cosmological quantity, including H_0, enters this calibration. The cosmological implications discussed in Section 4 therefore represent an independent consequence of the same empirically fixed scale.

Importantly, the calibration of $a_\star$ uses only galactic kinematic data and does not involve any cosmological observable.

4 COSMIC VOIDS AND THE HUBBLE TENSION

In this section we explore the cosmological implications of the effective saturation mechanism introduced above. We focus on late-time, low-density environments and show how the same effective transition structure responsible for flat galaxy rotation curves naturally leads to environment-dependent signatures in the locally inferred expansion rate.

The analysis is restricted to late-time phenomenology at low redshift. Early-universe physics and the global expansion history are assumed to remain unchanged.

It is important to clarify that the cosmological argument developed in this section does not follow from standard MOND phenomenology alone. In conventional MOND frameworks, the modified field equation is treated as a fundamental alteration of gravity at galactic scales, without generically implying an environment-dependent bias in the locally inferred Hubble parameter. Here, by contrast, the same bounded-response principle that governs galactic dynamics extends consistently to volumetric dilution in cosmic voids, linking rotation curves and Hubble inference within a single effective saturation mechanism.

Importantly, the saturation scale $a_\star$ entering the void analysis is fixed entirely by galaxy rotation curve data (Section 3). No cosmological parameter, including the Hubble constant, enters the determination of $a_\star$. The void analysis therefore does not tune H_0, but explores the consequences of a scale already fixed by independent galactic observations.

The background Friedmann evolution remains strictly unchanged. The mechanism discussed here does not alter the homogeneous expansion history, but affects the local kinematic inference of the expansion rate in strongly underdense environments.

Unlike conventional local-void explanations of the Hubble tension, which require a large-scale density fluctuation tuned to specific amplitudes and spatial profiles, the present mechanism follows directly from the same bounded-response principle that governs galactic dynamics. The effect is therefore not introduced ad hoc, but arises from a scale already constrained by independent data.

4.1 Local Expansion in Low-Density Environments

Measurements of the Hubble constant rely on local kinematic observables, including distance ladders, standard candles, and redshift–distance relations. These measurements implicitly assume that the locally inferred expansion rate is representative of the global cosmological background Riess (2022); Freedman (2021).

However, observations of large-scale structure show that the late-time Universe is strongly inhomogeneous. A substantial fraction of the cosmic volume is occupied by voids, characterized by matter densities well below the cosmic mean Keenan et al. (2013).

Within the effective framework introduced in Section 2, low-density environments probe the saturated regime of the effective response. In this regime, local kinematic observables no longer sample the same effective transition resolution as in high-density regions. As a result, expansion rates inferred from such environments need not coincide with the global average, even in the absence of any modification of early-time cosmology.

We therefore distinguish between a global expansion rate $H_{\rm global}$, defined by the homogeneous background constrained by early-universe observables Collaboration (2020), and an effective local expansion rate $H_{\rm loc}$, inferred from observations within a specific late-time environment.

4.2 Effective Expansion Rate in Voids

Cosmic voids represent the deepest and most extended low-density regions in the late Universe. Within voids, baryonic and total matter densities are strongly suppressed, and typical gravitational accelerations fall below the saturation scale $a_\star$ over large volumes Keenan et al. (2013).

In this regime, the effective description operates close to its low-density saturation limit. The relation between local kinematics and the underlying background expansion is therefore modified at the level of inference rather than at the level of the dynamical Friedmann equations. To leading order, this effect can be captured by an environment-dependent renormalization of the inferred expansion rate,

$$H_{\rm loc} = H_{\rm global}\left[1 + \delta_{\rm eff}\right], \quad \text{with} \quad \delta_{\rm eff} = \delta_\rho\, f\left(\frac{g_{\rm void}}{a_\star}\right), \tag{8}$$

where $\delta_\rho = 1 - \rho_{\rm void}/\rho_{\rm global}$ denotes the density contrast of the void, and $g_{\rm void}$ represents a characteristic gravitational acceleration scale associated with the void, estimated from its typical size and density contrast. This expression does not modify the background Friedmann equations. It parametrizes a bias in the kinematic inference of the expansion rate in underdense environments.

The function $f(x)$ is a dimensionless saturation function determined by the same bounded-response structure that governs galactic dynamics. In the deep saturated regime ($x \ll 1$), $f(x)$ approaches a finite asymptotic value $f_{\rm max}$, set by the maximal projection deficit allowed by the effective description. Independent constraints derived in Section 4.3 of Beau (2026a) imply $f_{\rm max} \lesssim 0.1$.

The saturation scale $a_\star$ entering Eq. (8) is fixed independently from galactic rotation curves (Section 3.4), and no cosmological parameter enters its determination.

For void density contrasts in the observationally inferred range $\delta_\rho \sim 0.2$–0.4 Keenan et al. (2013), and for $g_{\rm void} \ll a_\star$, Eq. (8) yields a characteristic fractional shift $\Delta H/H$ at the few-percent level. This magnitude is comparable to the observed local offset, without requiring any tuning of cosmological parameters.

The magnitude of the effect therefore follows directly from combining (i) the empirically determined scale $a_\star$, and (ii) independently measured void density contrasts.

The correction $\delta_{\rm eff}$ depends on the depth and spatial extent of the void, as well as on the empirically determined saturation scale $a_\star$. Denser or more weakly underdense regions remain close to the unsaturated regime and yield $\delta_{\rm eff} \approx 0$.

Importantly, this mechanism does not introduce any additional energy component and does not modify the background Friedmann equations. The global expansion history remains unchanged. Only the local kinematic inference of the expansion rate is affected.

Within this interpretation, the enhanced expansion inferred in cosmic voids reflects a reduced effective geometric constraint rather than an additional energy density. In regions where the density of contributing relational structure is low, the kinematic inference based on sparse mass distributions operates closer to its saturation limit, inducing a systematic environment-dependent bias in locally inferred kinematic relations.

4.3 Connection to the Hubble Tension

The Hubble tension refers to the discrepancy between early-time determinations of the Hubble constant, primarily inferred from cosmic microwave background measurements, and late-time local measurements based on the distance ladder Riess (2022); Collaboration (2020). Despite extensive investigation, no single explanation has yet achieved broad consensus Verde et al. (2019).

In the present framework, this discrepancy is interpreted as a manifestation of environment-dependent kinematic inference. Early-time measurements are anchored in the nearly homogeneous, high-density Universe, where the effective description operates in the unsaturated regime and recovers $H_{\rm global}$. By contrast, late-time distance-ladder measurements are performed in a highly structured Universe and may preferentially probe void-dominated environments.

If observers and calibrator systems reside in regions operating near the saturated regime, Eq. (8) predicts a systematic upward shift in the locally inferred expansion rate,

$$H_{\rm loc} > H_{\rm global}. \tag{9}$$

The magnitude of the shift depends on the depth and spatial extent of the surrounding underdensity.

Importantly, this mechanism does not alter the background Friedmann evolution. It does not introduce new relativistic degrees of freedom, nor does it modify early-time cosmology. The effect arises solely from the bounded-response structure governing late-time low-density environments.

The extent to which local underdensities alone can account for the full observed tension remains debated Shanks (2019); Kenworthy et al. (2019). The present framework does not posit a single finely tuned large-scale void. Instead, it provides a distinct mechanism in which the same empirically determined saturation scale $a_\star$ that governs galactic dynamics induces a systematic environment-dependent bias in late-time kinematic inference.

The proposal is therefore not a local-void cosmology, but a unified effective description in which both galaxy rotation curves and the Hubble offset arise from a common bounded-response principle.

Because the dynamics derives from an effective potential $\Phi_{\rm eff}$, the framework admits a conserved effective energy and a corresponding modified virial relation for stationary systems. This ensures that equilibrium disk galaxies remain dynamically stable in the transition regime and that the void mechanism does not introduce inconsistencies at galactic scales.

Early-universe inferences (CMB) probe a nearly homogeneous high-density regime well above the saturation threshold, and therefore recover $H_{\rm global}$.

4.4 Observable Signatures

The effective saturation framework leads to several testable observational signatures that distinguish it from both ΛCDM and purely local-void interpretations.

First, the locally inferred Hubble constant should correlate with the large-scale density environment. Measurements performed in or near deep voids are expected to yield systematically higher values of H_0 than those obtained in denser regions. This correlation should persist after controlling for known observational systematics and distance-calibration effects.

Second, the effect should exhibit a characteristic redshift dependence. At sufficiently large redshift, line-of-sight averaging over multiple environments suppresses the environmental bias. The inferred expansion rate should therefore converge toward $H_{\rm global}$, producing a gradual reduction of the offset with increasing survey depth.

Third, the framework predicts environment-dependent signatures in void-related observables, including weak lensing profiles and redshift drift measurements Cai et al. (2017). In particular, if the effective expansion rate inside voids is enhanced relative to the global value, the temporal evolution of redshift for sources embedded in deep underdensities should deviate from the homogeneous expectation in a systematic way.

More generally, the predicted shift depends not only on the local environment of the observer, but on the integrated low-density structure sampled along the line of sight. Both the location of the source and the geometry of intervening voids contribute to the inferred bias. This implies that samples with different sky coverage or different environmental selection criteria may yield systematically different H_0 estimates.

Importantly, the framework does not predict a unique numerical value for the offset in a given environment. Instead, once $a_\star$ is fixed independently from galactic rotation curves, the sign, scaling behavior, and environmental dependence of the effect are determined. This renders the proposal falsifiable with current and forthcoming large-scale-structure and distance-ladder data.

4.5 Predicted Amplitude and Redshift Dependence

A key feature of the present framework is that the magnitude of the locally inferred deviation from the global expansion rate is constrained by the saturation scale $a_\star$, which is fixed independently from galaxy rotation curves.

For representative underdensities characteristic of large cosmic voids on scales of a few hundred megaparsecs, and for $g_{\rm void} \ll a_\star$, the framework predicts a fractional enhancement

$$\frac{\Delta H}{H} \equiv \frac{H_{\rm loc} - H_{\rm global}}{H_{\rm global}} \sim \text{ a few percent}, \tag{10}$$

with the precise value controlled by the empirically determined scale $a_\star$ and by independently measured void density contrasts.

The effect is intrinsically redshift dependent. At higher redshift, large-scale structure is less developed, density contrasts are reduced, and line-of-sight averaging over multiple environments becomes increasingly efficient. Consequently, the locally inferred expansion rate is expected to converge progressively toward $H_{\rm global}$ with increasing redshift.

The framework therefore predicts a smooth decrease of $\Delta H/H$ as a function of survey depth. A persistent offset at high redshift, inconsistent with environmental averaging, would falsify the mechanism.

Conversely, deviations substantially larger than the percent-to-few-percent range, or the complete absence of any measurable environment-dependent trend once systematics are controlled, would directly challenge the proposed interpretation.

4.6 Interpretation of the Hubble Tension as an Environmental Effect

The framework developed here suggests that the Hubble tension does not reflect a change in the global expansion history. It reflects a limitation of homogeneous kinematic inference when applied to a strongly inhomogeneous late-time Universe.

Early-universe determinations of H_0, anchored in a nearly homogeneous background, recover the global expansion rate. By contrast, late-time local measurements probe structured environments in which the effective transition resolution operates in a low-density saturated regime.

In this context, the discrepancy between early- and late-time determinations of H_0 arises from a mismatch between global averaging and local inference. Measurements performed in void-dominated regions sample an effective response that no longer scales linearly with decreasing density, leading to a systematically enhanced locally inferred expansion rate.

From this perspective, the Hubble tension is not an anomaly requiring additional dark components or early-time modifications. It is a diagnostic of gravitational and kinematic inference in diffuse environments. This interpretation naturally predicts weak but systematic correlations between inferred values of H_0 and large-scale environmental indicators such as void depth or filament membership.

4.7 Summary

In this section we have shown that the same effective saturation mechanism that accounts for galaxy rotation curves naturally extends to late-time, low-density cosmological environments.

When the characteristic gravitational field falls below the empirically determined scale $a_\star$, the effective description operates in a bounded-response regime. In cosmic voids, this regime induces an environment-dependent bias in the locally inferred expansion rate, while leaving the global background dynamics unchanged.

The mechanism therefore provides a structural link between galactic kinematics and the Hubble tension: both phenomena arise from the same transition scale, independently fixed by rotation curve data.

The predicted amplitude of the effect is constrained to the percent-to-few-percent level, depends on independently measured void density contrasts, and decreases with increasing redshift due to environmental averaging. These features render the proposal observationally testable.

In the next section, we examine the robustness of these results, discuss degeneracies with standard cosmological effects, and clarify the domain of validity of the effective description.

5 ROBUSTNESS, DEGENERACIES, AND LIMITS

In this section we assess the robustness of the effective saturation framework, its potential degeneracies with standard astrophysical and cosmological effects, and the limits of its domain of applicability. This analysis is essential to determine whether the proposed mechanism constitutes a genuine explanatory alternative or merely a reparameterization of existing descriptions.

5.1 Baryonic Uncertainties

The predictions of the effective framework depend explicitly on the observed baryonic mass distribution. Uncertainties in stellar mass-to-light ratios, gas content, and distance estimates therefore propagate into the inferred rotation curves. Such uncertainties are well documented and represent a generic limitation of baryon-based dynamical modeling Lelli et al. (2016).

We have tested the sensitivity of the effective fits to reasonable variations in stellar mass normalization. While moderate shifts in mass-to-light ratio affect the detailed shape of the inner rotation curves, they do not eliminate the emergence of a low-acceleration saturation regime at large radii. In particular, the appearance of asymptotically flat rotation curves remains robust across the allowed baryonic parameter range, consistent with empirical scaling relations McGaugh et al. (2016).

At cosmological scales, baryonic uncertainties primarily affect the mapping between large-scale structure and local observational samples. They do not remove the qualitative distinction between dense environments, which probe an unsaturated effective regime, and cosmic voids, which sample the saturated regime most strongly Keenan et al. (2013).

5.2 Conceptual Position within Modified-Gravity and Inhomogeneous Frameworks

The phenomenology discussed in this work exhibits partial overlap with acceleration-based approaches such as Modified Newtonian Dynamics (MOND), and the cosmological discussion involves large-scale underdensities. It is therefore important to clarify the conceptual position of the present framework within this broader landscape.

5.2.0.1 Relation to MOND-like phenomenology. At galactic accelerations below $a_\star$, Eq. (1) becomes formally equivalent to quasi-linear MOND in the weak-field regime. Both approaches identify a characteristic acceleration scale below which Newtonian expectations break down.

The distinction is structural rather than mathematical. In MOND, the acceleration scale a_0 is introduced as a fundamental modification of the gravitational law. In the present framework, the same scaling behavior arises as the low-density limit of a bounded-response principle. The scale $a_\star$ characterizes the onset of saturation in the effective description, not a new force law.

Crucially, the extension to cosmological inference does not follow from standard MOND formulations. Here, both galactic dynamics and the environment-dependent Hubble offset derive from the same empirically calibrated scale $a_\star$, linking two regimes that are usually treated independently.

5.2.0.2 Distinction from inhomogeneous cosmologies. Models based on large-scale metric inhomogeneities, such as Lemaître–Tolman–Bondi (LTB) constructions, attempt to account for late-time discrepancies by modifying the background geometry itself. In such scenarios, the Hubble parameter becomes position dependent because the spacetime metric deviates from the homogeneous Friedmann–Lemaître–Robertson–Walker solution.

The present framework does not modify the background metric and does not introduce a position-dependent Friedmann solution. The global expansion history remains that of standard cosmology. The predicted offset arises instead from an environment-dependent modification of the effective kinematic inference once local accelerations fall below the saturation scale $a_\star$.

The mechanism therefore does not require a special observer location or a finely tuned void profile. The shift in the locally inferred expansion rate scales with the surrounding density contrast according to Eq. (8), while preserving global homogeneity at the level of the background solution.

5.2.0.3 Structural distinction. The unifying feature of the framework is the existence of a universal bound on effective response. This bound is not environment dependent. What varies across environments is the density of relations contributing to the effective projection.

Dense systems remain in the unsaturated regime, while extended low-density regions approach saturation. The resulting correlation between galactic kinematics and late-time Hubble inference is therefore not a reparameterization of dark sector physics nor a modification of the cosmological metric, but a manifestation of a single bounded-response structure operating across scales.

5.3 Degeneracies with Standard Astrophysical Effects

Any proposed alternative mechanism must be carefully examined against established astrophysical and cosmological effects within the ΛCDM framework. We therefore distinguish between degeneracies at galactic and cosmological scales.

5.3.0.1 Galactic scales. Within dark matter halo models, baryonic feedback processes and baryon–halo coupling have been shown to modify inner halo profiles and reproduce certain features of flattened rotation curves, particularly in low-mass systems Di Cintio et al. (2014). Such models typically introduce galaxy-dependent halo parameters and rely on a coupling between baryonic surface density and halo response.

By contrast, the effective saturation framework introduces no halo degrees of freedom and involves a single universal acceleration scale $a_\star$. All galaxy-to-galaxy diversity arises solely from the observed baryonic mass distribution and from the fitted stellar mass-to-light ratio. The resulting scaling relations follow directly from the structure of Eq. (1), without requiring halo-by-halo tuning.

A potential degeneracy therefore exists at the level of rotation curve fitting accuracy, but the underlying parameter structure differs qualitatively. Detailed comparisons of residual patterns and scaling relations provide a discriminant between a universal saturation mechanism and feedback-regulated halo models.

5.3.0.2 Cosmological scales. At low redshift, local inhomogeneities, peculiar velocities, and sample variance are known to affect distance-ladder determinations of the Hubble constant Freedman (2021). Large-scale underdensities have also been explored as partial explanations of the Hubble tension Shanks (2019); Kenworthy et al. (2019). In most analyses, however, these effects are treated as perturbations of the background expansion within standard gravity.

The present framework differs in a crucial respect. The background Friedmann expansion remains unmodified. The predicted offset arises not from a change in the global dynamics, but from an environment-dependent modification of the effective kinematic inference once the local acceleration falls below the saturation scale $a_\star$.

In particular, the mechanism does not rely on a finely tuned radially symmetric void profile or on a large-scale violation of homogeneity. Instead, it predicts a systematic scaling of the inferred expansion rate with the local density contrast, as encoded in Eq. (8). This scaling is tied to the same universal parameter $a_\star$ determined independently from galaxy rotation curves.

5.3.0.3 Conceptual distinction. A central ingredient of the framework is the existence of a universal bound on the effective response of the geometric description. This bound is not environment dependent. What varies across environments is the density of relations contributing to the effective projection.

Dense systems concentrate many effective relations within limited volumes, keeping the dynamics close to the unsaturated regime. By contrast, extended low-density regions distribute the same fixed resolution over larger volumes, pushing the effective description closer to saturation. The resulting enhancement of the locally inferred expansion rate therefore reflects how different environments sample a fixed bounded-response structure, rather than a modification of gravity or a reparameterization of dark sector physics.

This distinction cannot be absorbed into standard astrophysical degeneracies. It predicts correlated galactic and cosmological signatures tied to a single universal scale, providing a clear avenue for empirical discrimination.

5.4 Environmental Selection Effects

Local measurements of the Hubble constant are not uniformly distributed across cosmic environments. Distance ladder calibrators and standard candles are preferentially observed in specific regions of the large-scale structure Riess (2022).

This selection effect plays a central role in the present framework. If observational strategies preferentially sample void-dominated or underdense regions, the inferred expansion rate is expected to be systematically enhanced relative to the global value.

Importantly, this effect is not introduced as an ad hoc assumption. It constitutes a direct and testable prediction. Future analyses correlating inferred values of H_0 with void catalogs and density reconstructions can directly assess the magnitude of the effect, as already explored in the context of local underdensity scenarios Shanks (2019); Kenworthy et al. (2019).

5.5 Range of Validity

The effective saturation framework is not intended to apply universally at all scales and epochs. Its domain of validity is restricted to late-time, low-density environments.

In high-density regimes, including the early Universe, galaxy interiors, and the Solar System, the effective description reduces to standard Newtonian and relativistic behavior by construction. No deviations from established physics are expected in these contexts, consistent with precision tests of gravity Will (2014).

At sufficiently large redshift, line-of-sight averaging over multiple environments suppresses the saturation signature. The framework therefore predicts convergence toward standard cosmological behavior at high redshift, in agreement with current observations Collaboration (2020).

5.6 Falsifiability

The effective saturation framework makes a number of concrete, observationally testable predictions. Because the same universal scale $a_\star$ governs both galactic dynamics and late-time kinematic inference, the model can be falsified at multiple, independent levels.

5.6.0.1 (i) Absence of environment dependence in H_0. The framework predicts a correlation between the locally inferred Hubble parameter and the surrounding large-scale density contrast, as expressed in Eq. (8). If environment-stratified analyses of distance-ladder measurements reveal no statistically significant trend between inferred H_0 and void proximity or density contrast, the proposed mechanism would be strongly disfavored.

5.6.0.2 (ii) Redshift independence of the local offset. The predicted deviation $\Delta H/H$ is expected to decrease with increasing redshift due to environmental averaging. A persistent offset of comparable magnitude at higher redshift, inconsistent with line-of-sight averaging, would contradict the core mechanism.

5.6.0.3 (iii) Inconsistency of the universal scale. The acceleration scale $a_\star$ is fixed from galaxy rotation curves. If independent cosmological analyses required a significantly different effective scale to reproduce any environment-dependent expansion signal, the internal consistency of the framework would be violated.

5.6.0.4 (iv) Breakdown of galactic saturation behavior. The model predicts systematic low-acceleration behavior tied to the same universal scale across high- and low-surface-density galaxies. A statistically significant population of extremely diffuse systems that cannot be reconciled with a single $a_\star$ would invalidate the bounded-response interpretation.

These criteria render the framework empirically vulnerable. It does not merely accommodate the observed anomalies, but constrains their amplitude, scaling, and redshift dependence. Failure of any of the above conditions would require revision or rejection of the proposed mechanism.

5.7 Summary

The effective saturation framework is robust against reasonable baryonic uncertainties and cannot be trivially absorbed into existing astrophysical or cosmological effects. Its explanatory power arises from the use of a single universal scale and from its explicit environmental dependence.

The framework is limited in scope but sharply testable. Its validity hinges on future observational probes of low-density environments, making it a falsifiable proposal rather than a flexible phenomenological fit.

6 PREDICTIONS AND OBSERVATIONAL TESTS

The effective saturation framework introduced in this work leads to a set of distinct and testable predictions. These predictions follow directly from the environmental dependence of the effective transition resolution and do not rely on additional assumptions or parameter tuning.

6.1 Environment-Dependent Hubble Measurements

A primary prediction of the framework is a correlation between the locally inferred Hubble constant and the surrounding large-scale environment.

Observers and standard candles located in or near deep cosmic voids are expected to yield systematically higher inferred values of H_0 than those located in denser regions. Conversely, measurements anchored in overdense environments should converge more closely toward the global expansion rate H_{global}.

This prediction can be tested by cross-correlating existing and forthcoming H_0 measurements with void catalogs and density reconstructions derived from large-scale galaxy surveys, as already explored in the context of local underdensity scenarios Keenan et al. (2013); Shanks (2019); Kenworthy et al. (2019). A statistically significant null result would strongly disfavor the effective saturation mechanism.

In particular, the framework predicts a differential signal: measurements performed in environments of comparable redshift but contrasting large-scale density should exhibit systematically different inferred values of H_0.

6.2 Redshift Dependence of the Hubble Offset

The environmental bias predicted by the framework is inherently scale dependent.

At sufficiently low redshift, local measurements are sensitive to individual large-scale structures and directly probe the saturated regime. At increasing redshift, line-of-sight averaging over multiple environments progressively suppresses the bias.

As a result, the inferred expansion rate is predicted to converge smoothly toward $H_{\rm global}$ with increasing redshift.

This behavior provides a clear observational signature. Measurements of the Hubble constant performed over different redshift ranges should exhibit a systematic trend, with the largest deviations appearing at the lowest redshifts, followed by a gradual convergence at higher redshift Verde et al. (2019).

An important quantitative consequence of the framework is that the expected deviation remains at the percent level. Once the saturation scale $a_\star$ is fixed by galactic dynamics, the fractional offset in the inferred expansion rate is predicted to peak at intermediate redshift and to remain negligible at both very low and sufficiently high redshift.

The peak is expected when the typical line-of-sight length becomes comparable to the characteristic scale of large voids, before full environmental averaging sets in.

Such deviations fall within the sensitivity range of forthcoming large-scale surveys, including DESI and *Euclid*. Observed deviations significantly larger or smaller than this level would directly challenge the proposed mechanism.

6.3 Primordial Saturation and Direct Collapse

The redshift dependence discussed above acquires particular significance in the primordial regime. As introduced in Section 2.4, the operational saturation scale

$$a_\star(z) \sim \kappa \frac{b_\chi}{\ell(z)} \tag{11}$$

depends inversely on the descriptive resolution $\ell(z)$. At very high redshift, the projected state remains spectrally constrained, so that $\ell(z)$ cannot decrease arbitrarily while preserving smooth quasi-injective updates.

Consider a collapsing primordial gas cloud of mass M and radius R. The characteristic update demand is set by the free-fall timescale,

$$D(R) \sim \frac{1}{t_{ff}} \sim \sqrt{\frac{GM}{R^3}}. \tag{12}$$

Fragmentation into stellar units requires that the characteristic fragmentation scale λ_{frag} remain larger than the minimal projectable scale $\ell(z)$. When $\lambda_{frag} < \ell(z)$, the projection cannot resolve a multi-body stellar configuration.

In this regime, admissibility can only be restored through a saturated reconfiguration channel. Direct collapse into a compact seed then becomes the structurally available outcome. This yields an implicit redshift-dependent critical scale $M_{crit}(z)$ defined by the condition that the required fragmentation scale equals $\ell(z)$. Massive halos above this threshold naturally enter a direct-collapse regime without requiring additional exotic physics.

6.4 Scaling Analysis of the Projective Gap

The inevitability of saturation can be established through a simple scaling argument. Let the update demand scale as

$$D(R) \propto R^{-3/2}, \tag{13}$$

as dictated by the free-fall timescale.

The projective capacity is limited by the maximal admissible throughput under the invariant bound b_χ. Denote by $C(R)$ the total admissible update capacity within a region of size R. Without assuming a specific geometric postulate, we parametrize

$$C(R) \propto R^n, \tag{14}$$

where $n \geq 0$ encodes the effective growth of independent relational channels permitted by the mapping Π at the epoch considered.

The projective stress ratio behaves as

$$\Xi(R) = \frac{D(R)}{C(R)} \propto R^{-(3/2+n)}. \tag{15}$$

For any physically reasonable $n \geq 0$, the ratio diverges as $R \to 0$. This guarantees the existence of a finite contraction scale R_{sat} at which $\Xi \sim 1$. Beyond this point, the demanded update rate exceeds the admissible projective throughput.

The result is robust with respect to the precise value of n. Even if capacity scaled volumetrically ($n = 3$), the divergence would remain unavoidable. Saturation during sufficiently deep collapse is therefore a structural feature of the framework rather than a model-dependent artifact.

6.5 Little Red Dots as Early Saturation Signatures

Recent JWST observations have revealed a population of compact, luminous, high-redshift sources often referred to as "Little Red Dots" (LRDs) Pacucci et al. (2025); Rusakov et al. (2026). These objects have been interpreted as rapidly growing massive black holes, possibly embedded in dense ionized cocoons, implying efficient early mass assembly at redshifts $z \gtrsim 6$.

Within the present framework, such sources may be interpreted as manifestations of primordial projective saturation. Two non-exclusive realizations are possible. First, halos exceeding $M_{crit}(z)$ may undergo genuine direct collapse, producing massive compact seeds at early epochs. Second, a strongly constrained relational configuration may generate an enhanced effective convergence through the emergent contribution κ_χ introduced in Section 11.10 of the White Paper, without requiring proportionally large baryonic mass.

In both cases, the underlying mechanism is identical: a resolution gap between the demanded fragmentation scale and the minimal projectable scale $\ell(z)$. The observational signature is therefore not the presence of an exotic component, but the structural response of the projection under extreme compression.

Discriminating between a physical direct-collapse black hole and a purely projective overmass configuration requires combined constraints from variability, spectral diagnostics, and dynamical mass estimates. The framework predicts that the abundance of such compact early objects should track the redshift evolution of the effective resolution scale rather than solely the mass accretion history of halos.

6.6 Void-Specific Redshift Drift

The effective saturation mechanism also affects the temporal evolution of redshift for sources embedded in low-density environments.

In the standard cosmological framework, the redshift drift depends solely on the global expansion history. In the present framework, sources located in deep voids experience a modified effective transition rate, leading to a small but systematic deviation in the expected redshift drift signal.

Future high-precision spectroscopic experiments may be able to detect this effect by comparing sources located in voids with those in denser regions, complementing existing redshift-drift proposals Cai et al. (2017). The effect is predicted to be suppressed at high redshift, where environmental averaging dominates.

6.7 Weak Lensing Signatures in Voids

Because the effective framework modifies the inferred gravitational response in low-density regions, it also impacts weak gravitational lensing by cosmic voids.

The model predicts subtle but systematic deviations in void lensing profiles relative to standard expectations. In particular, the effective saturation leads to a systematically altered lensing efficiency compared to what would be inferred from baryonic matter alone, without invoking additional dark components.

Void lensing measurements therefore provide an independent probe of the framework, complementary to kinematic tests, and have been identified as sensitive diagnostics of gravitational behavior in underdense regions Cai et al. (2017).

6.8 Predictions for Ultra-Diffuse Galaxies

Ultra-diffuse galaxies constitute extreme low-surface-density systems and probe the saturated regime across most of their spatial extent van Dokkum (2016).

Quantitatively, in the saturated regime ($g_{\rm N} \ll a_\star$), the effective framework predicts that the asymptotic circular velocity is bounded by the finite response of the effective gravitational field. To leading order, this implies

$$v_\infty \sim \sqrt{R\, a_\star}\, \left[1 - O(\delta_\rho)\right], \tag{16}$$

where R is the characteristic baryonic scale length of the system and $\delta_\rho = 1 - \rho_{\rm void}/\rho_{\rm global}$ is the local underdensity contrast. For ultra-diffuse galaxies in void environments ($\delta_\rho \approx 0.3$–0.6 and $R \sim 3$–5 kpc), this yields typical asymptotic velocities $v_\infty \lesssim 10$–$15\ {\rm km\,s^{-1}}$.

This prediction differs qualitatively from MOND, which enforces a universal relation $v_\infty \propto (M_{\rm baryon} a_0)^{1/4}$ independent of environment. In the present framework, the asymptotic velocity depends explicitly on the large-scale environment through δ_ρ, leading to systematically lower velocities for ultra-diffuse galaxies in voids than for systems of comparable baryonic mass in denser regions.

Observations reveal a wide diversity of inferred dynamical mass content among ultra-diffuse galaxies, ranging from apparently dark-matter-rich systems to galaxies consistent with little or no dark matter van Dokkum (2019); Mancera Piña (2019). Within the present framework, this diversity arises naturally from differences in how deeply individual systems probe the saturated regime.

In ultra-diffuse galaxies, the apparent mass discrepancy inferred from kinematics should therefore not be interpreted as evidence for an extended distribution of unseen matter. It reflects a geometric threshold effect. The baryonic configuration probes a regime in which further density dilution no longer increases the effective dynamical response.

As a result, the inferred dynamical mass tracks the onset of saturation rather than the presence of an additional gravitating component. The apparent "missing mass" is thus an emergent consequence of limited descriptive resolution in low-density systems.

This interpretation leads to a falsifiable prediction. If the inferred mass discrepancy originates from geometric saturation rather than from a material mass component, independent mass tracers should not reveal correspondingly extended matter distributions.

In particular, weak lensing signals, satellite dynamics, and stellar velocity dispersion profiles are expected to systematically underpredict the dynamical mass inferred from kinematic measurements in the deepest saturation regime. A positive detection of massive, spatially extended halos in ultra-diffuse galaxies would directly falsify the present framework.

Because ultra-diffuse galaxies probe the deepest saturation regime, they provide a particularly sensitive test of the universality of the saturation scale $a_\star$.

6.9 Summary

The effective saturation framework yields a coherent and tightly constrained set of predictions across both galactic and cosmological scales.

All predictions arise from environmental dependence and involve no additional free parameters beyond the saturation scale $a_\star$ already fixed by rotation curve data.

The framework is therefore falsifiable with current or near-future observational capabilities. Confirmation or rejection of these signatures will decisively determine whether the proposed mechanism captures a genuine aspect of low-density gravitational phenomenology.

7 CONCLUSION

We have examined whether two long-standing observational anomalies, namely flat galaxy rotation curves and the Hubble constant tension, may originate from a common effective mechanism operating in low-density environments.

Using a minimal phenomenological framework characterized by a single saturation scale, we have shown that baryonic matter alone can reproduce the main features of observed galaxy rotation curves across a wide range of systems, without halo-by-halo tuning or the introduction of dark matter particles within this effective description. This conclusion is supported by direct comparisons with representative rotation curves from the SPARC sample.

The same effective mechanism extends naturally to cosmological low-density regions. Cosmic voids, which probe the deepest saturation regime, emerge as key laboratories for this framework. In these environments, the locally inferred expansion rate is predicted to deviate systematically from the global value inferred from early-universe observables, while the background cosmological evolution remains unchanged.

Crucially, the framework does not modify early-time physics, does not alter the global expansion history, and does not claim universal validity across all regimes. Quantities such as gravitational acceleration and the Hubble parameter are interpreted as effective inference rates, not as fundamental dynamical fields.

A distinctive strength of the approach lies in its cross-sector falsifiability. Because the same saturation scale governs both galactic dynamics and environment-dependent expansion inference, observational constraints obtained in one sector directly restrict the allowed phenomenology in the other. In particular, measurements of ultra-diffuse galaxy kinematics, which probe the deepest saturation regime at galactic scales, place quantitative bounds on the magnitude of any environment-dependent offset in the inferred Hubble constant. Conversely, the absence of such an offset in cosmological observations would falsify the interpretation of galactic low-acceleration behavior as a saturation effect.

The strength of the framework thus lies not in fitting individual anomalies in isolation, but in tying together galactic and cosmological phenomenology through a single effective scale and a shared environmental dependence. This coherence sharply limits the available parameter space and prevents independent adjustment of the two sectors.

We have identified several observational tests capable of confirming or excluding the framework. These include correlations between inferred values of H_0 and large-scale environment, redshift-dependent suppression of the local offset, void-specific redshift drift, weak lensing signatures, and the dynamics of ultra-diffuse galaxies.

Taken together, these results suggest that galaxy rotation curves and the Hubble tension may reflect a shared low-density phenomenology rather than independent failures of the standard cosmological model. Whether this effective description captures a genuine aspect of gravitational inference or represents an intermediate phenomenological layer will ultimately be decided by targeted observational tests in the coming years.

DATA AVAILABILITY

The galaxy rotation-curve data used in this work are drawn from the publicly available SPARC database Lelli et al. (2016).

The numerical code used to generate the galaxy rotation-curve figures and associated diagnostics is openly available and archived with a persistent identifier. The version used in this work is available via Zenodo at `https://doi.org/10.5281/zenodo.18462369` Beau (2026d).

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